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(1) Research Program/Project/Study Title: Effectiveness on the Use of ICT Instructional Materials in Classes Among BSCS Students of CapSU	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation Capiz State University (All campuses)			
Municipality/City:		Roxas	
Province:		Capiz	
Region:		VI	
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Education		(8) Discipline/Field of Specialization Computer Education	

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Access to education is necessary at all times and places because it is a lifelong process (Kaplan. 2016). Since there is a constant and everlasting need to access information due to the information explosion, IT plays a crucial role in guaranteeing that education addresses the needs of a wide range of learners. However, the rate of change brought about by new technologies has altered how people live, work, and play across the globe. Emerging and current technologies are putting traditional teaching and learning methods to the test, as well as the way that education is managed. Efficient communication combined with sufficient access to IT in the home, office, or classroom makes learning a truly lifetime endeavor. The speed at which technology is developing necessitates regular observation and evaluation of the teaching process. Even to a certain extent, the home is currently becoming a laboratory where the whole family interact through mobile phones and gadgets.

Computer has continued to play significant role in the development of any society. According to Attach (2003) computer is an electronic and automatic machine that accepts data through the central processing units which cannot be achieved through in-person contact between employer and employees. It requires details and specific instruction or discussion of the messages or information. That is why, computer is taught to people to enable them think creatively and constructively in computer terms and to apply it in the analysis of data. But, in spite of the important place of computer in the educational system, students in schools continually register fair performances (Obodo 2002). Hence, IMs need updating, that is if the real intention is to enhance teaching-learning process. Further, Betiku (2002) has the opinion that the fair performance may be due to methodology of teaching, attitude of teachers or lack of interest by students. But the teachers' favorable response to the challenge were limited by several factors. It was due to teachers' commitment to better education was the driving force for them to keep revising their IMs despite the inadequacies.

As a known fact, the problems of using instructional materials are the shortage of materials, preoccupation amongst the teachers in other school works, lack of fund for the purchase of instructional materials, and time (Dhakal, 2020). Another is the poor ICT Infrastructure is also another important issues and challenge in successful implementation of ICT in education. Infrastructure such as hardware, software, internet access, and similar other resources which are required to integrate technology in teaching and learning are not sufficient. With the situation and the demand of the workplace, teachers are deemed to create instructional materials that would make learning motivating and challenging to students. Instructional materials serve as a channel between the teacher and the students in delivering instructions. Instructional materials are highly important for teaching. Teachers rely on instructional materials in every aspect of teaching. They need material for background information on the subject they are teaching. Having seen the important of computer as an inevitable tool for human and technological development, it is necessary to look into the use of instructional materials for effective teaching and learning. Because of these conflicting views on the use of instructional materials for effective teaching and learning the conduct a survey on the effectiveness use of instructional materials in teaching-learning in the Bachelor of Science in Computer Science was conceptualized.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

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In order to ascertain the effectiveness of the use of ICT Instructional Materials in classes of college students of the Computer Studies Department, Capiz State University, Dayao Satellite College, the following procedures will be employed.

- (1) Computation and stratification of sample size
The respondents of this study will be the one hundred eighty-eight (188) students from the Computer Studies Department, randomly selected first-year to fourth-year college students.
- (2) Formulation of researcher-made questionnaires
The research instrument that will be used is the researcher-made survey instrument that will be validated by the panel of experts. For the reliability of the instrument, thirty (30) students will participate in supplying the needed data. The Cronbach Alpha will show results that signify the reliability of the instrument.
- (3) Pilot testing
The instruments will be pilot tested among randomly selected thirty (30) randomly selected first-year to fourth-year college students. who are not part of the sample group. Pilot testing will pinpoint the problem areas of the survey questionnaires.
- (4) Item analysis
The index of difficulty and index of discrimination will be used whether to accept or discard some items in the test.
- (5) Conduct of study
Descriptive-correlational research will be conducted.
- (6) Data analyses and interpretation
The statistical analyses to be used, like mean, percentage, t-test, and Analysis of Variance (ANOVA) for significant differences between independent and dependent variables, were used.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The main purpose of the study is to determine the effectiveness of the Use of ICT Instructional Materials in Classes Among BSCS Students of CapSU.

Specifically, it will answer the following;

1. What is the socio-demographic profile of the respondents when grouped according to age, sex, place of residence, year level and monthly family income?
2. What is the level of effectiveness on the use of instructional materials in terms of cognitive, affective and psychomotor learning?
3. What is the degree of teaching effectiveness of instructors as aided by the use of instructional materials?
4. Is there a significant difference on the effectiveness of ICT instructional materials in cognitive, affective and psychomotor learning of students and their profile?
5. Is there a significant relationship on the effectiveness of ICT instructional materials in cognitive, affective and psychomotor learning of students and their profile?

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Research publications evaluating the efficacy of instructional materials in ICT education have significantly increased in tandem with the growing usage of technologies in education. These publications represent a substantial body of untapped research that offers important insights by analyzing the quantitative findings of multiple empirical studies. Various scholars have already conducted analysis in an effort to present a comprehensive

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assessment of the effectiveness of instant messaging (IMs) in facilitating learning effectiveness.

Chauhan's (2017) study uncovered prior research that indicates technology can be a potent aid for students' successful learning provided it is fully integrated into pedagogy or instruction. Actually, a study that examined the integration of the quantitative results of 122 peer-reviewed scholarly articles that assessed the influence of technology on students' learning effectiveness was carried out. The findings verified that the impact of technology on students' learning effectiveness is moderate. The results supported learning effectiveness when the study examined the effect sizes of moderating variables such domain subject, application kind, intervention length, and learning environment.

In an effort to boost academic performance, some school districts in nations like the USA have recently committed a sizeable portion of their yearly budgets to the use of technology in the classroom. Given that the majority of public schools have access to computers with Internet connections, they are in a far better position to integrate technology into the classroom. Few schools have parents, teachers, administrators, and other stakeholders all sharing a commitment to technology (Lim, Zhao, Tondeur, Chai, & Tsai, 2013). According to Broekhuizen (2016), there could be several factors that contribute to it, including the training and preparation of teachers, the impact of technology on school culture, and worries about the accessibility of technology at home that could exacerbate inequalities among students from different socioeconomic backgrounds.

Additionally, it is imperative that educational establishments comprehend the principles and guidelines pertaining to the efficient incorporation of technology in the classroom. Schaffhauser (2016) proposed that efforts should be made to improve classroom instruction in order to better employ technology for learning. Since it will enable them to think about how they explicitly use technology in the classroom, teachers should be included in the creation of a school-based technology plan. The cornerstones of integrating technology in education are close collaboration between the different stakeholders within the school community and a shared understanding of technology usage.

The study's goal of Amos et al., (2022) was to investigate how instructional materials affected biology instruction and learning in Ghana's Central Region's Colleges of Education. The study found that effective lesson delivery has a significant impact on teaching and learning environments, and that students have access to a wealth of information to support their education and learn about the proper use of instructional materials. It is advised that professional development programs be set up so that biology tutors at the various colleges can become proficient in using pertinent teaching resources to evaluate students during biology classes. Additionally, it is advised that universities, the National Teaching Council (NTC), and the Ghana Tertiary Education Commission (G-TEC) work together with the colleges to plan workshops and seminars where tutors teaching biology can refresh their knowledge on how to present lessons using digital instructional materials.

As UMAR et al., (2019) expound that usefulness of instructional materials in computer science instruction was examined in this study conducted at education colleges. Instructional materials were used to teach fifty students (the experimental group) and fifty students (the control group) were not given any. At the 0.05 significant level, the hypothesis was tested using an independent t-test, and the research question was examined using the Pearson product moment correlation coefficient at that level. According to the study, using instructional materials improved students' conceptual understanding and led to high academic performance in general. Pupils who received instruction using instructional materials outperformed those who did not receive them. There were recommendations on how to encourage the use of educational resources in computer instruction in order to raise the academic achievement of computer students.

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(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

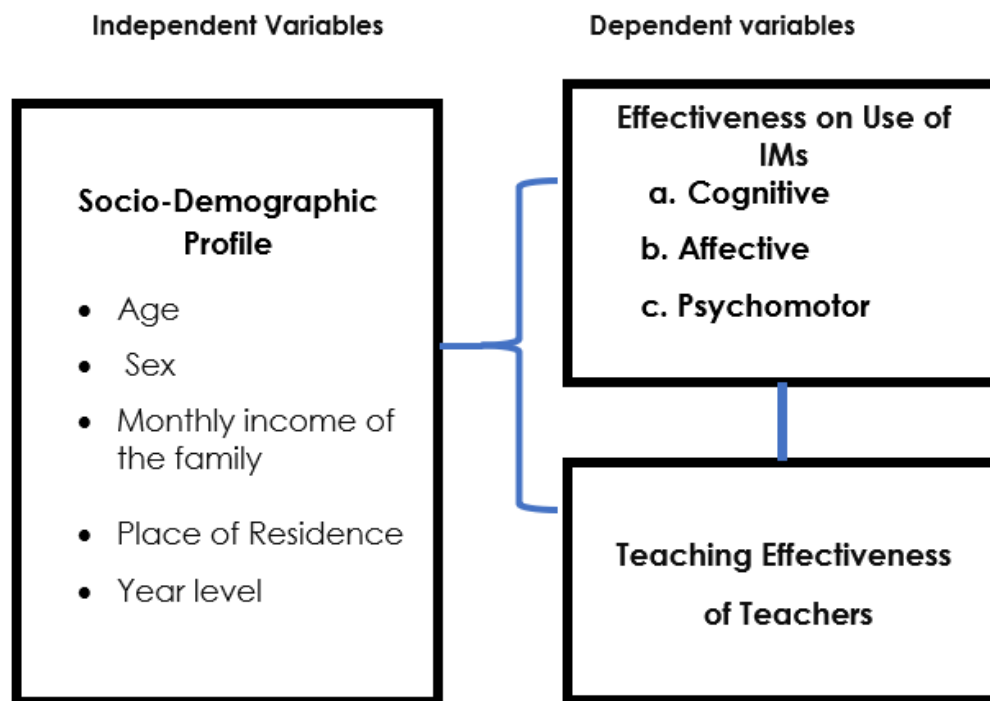


Figure 1. Schematic diagram showing the relationship between the independent variables and dependent variables

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

The dependent variable in this study will be the effectiveness of the use of ICT Instructional Materials, and the independent variable will be the use of ICT instructional materials

A 30-item researcher-made test will be used to assess students' effectiveness of ICT instructional materials in cognitive, affective, and psychomotor learning of students and their profile. Each question will be in the form of a multiple choice. Prior to the study's conduct, the instrument will undergo content validation by experts in the field and will be pilot-tested among thirty (30) first-year to fourth-year college students who are not part of the sample group. The index of difficulty and index of discrimination will be used to determine whether to accept or discard some items in the test. Statistical description will be utilized to analyze data.

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The questionnaire will be a 5-point Likert scale covering students' cognitive, affective, and psychomotor learning.

Scoring Responses

The scoring of the variable is based on the scoring guide.



Score	Categori es	Scoring Interval	Description
5	Very High /Extent	4.21–5.00	If you find the level of effectiveness on the use of IMs to enhance learning of students is manifested excessively or more than necessary
4	High/ High Extent	3.41–4.20	If you find the level of effectiveness on the use of IMs to enhance learning of students is manifested slightly excessive or just necessary.
3	Fairly High/Mod erate Extent	2.61–3.40	If you find the level of effectiveness on the use of IMs to enhance learning of students is manifested <u>justly</u> or equally necessary rather than adequate or essential
2	Low/ Extent	1.81-2.60	If you find the level of effectiveness on the use of IMs to enhance learning of students manifestly fell short of what is <u>required</u> or essential
1	Very Low /Extent	1.00–1.80	If you find the level of effectiveness on the use of IMs to enhance learning of students manifestly well short of what is required or essential.



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To determine the effectiveness of the use of ICT instructional materials in classes among BSCS students of CapSU, survey-correlational research will be conducted.

The respondents of this study will be the one hundred eighty-eight (188) students from the Computer Studies Department, Capiz State University, Dayao Satellite College, randomly selected first-year to fourth-year college students. The Cochran formula will be used in the computation of the sample size.

The statistical analyses to be used are mean, percentage, t-test, and Analysis of Variance (ANOVA).

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	2024											
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1. Proposal Writing												
2. Proposal Presentation												
3. Procurement of materials												
4. Field Work												
5. Data gathering												
6. Data Analysis												
7. Research paper writing												
8. Colloquim/Defense												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

Provide valuable insights for enhancing teaching and learning practices in the field, Optimizing the use of ICT materials in BSCS classes, including suggestions for improvement and areas for further research.

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Transferring the effectiveness of using ICT instructional materials in classes among Bachelor of Science in Computer Science (BSCS) students into extension activities involves

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extending the benefits and lessons learned from the classroom to broader contexts, such as workshops, seminars, community outreach programs, and collaborative projects.

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

The results of this study may show that the effective use of ICT instructional materials in BSCS classes can have far-reaching positive impacts on individuals, educational institutions, industries, and society as a whole. Clientele are BSCS Students, Faculty Members, Educational Institutions, Employers and Industry, Government and Policy Makers, and Society at Large.

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

	Counterpart Funding	Other Funding	TOTAL
I. Personal Services			
Salaries/ Honorarium			
II. Maintenance and Other Operating Expenses			
A. Direct Cost			
Travel Expenses (Local)			P 3,000
Supplies and Materials			
Raw Materials			
Tools			
Facility and Equipment Rental Expenses			
Communication Expenses			P 3,000
Documentation Printing, Photocopying Expenses			P 7,000
B. Indirect Cost			
Utilities/Miscellaneous			P 3,000
GRAND TOTAL			P 16,000.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from

Ahmed, S., & Alam, F. (2020). Impact of Information and Communication Technology (ICT) in Education: A Review of the Literature. *International Journal of Research in Education and Science (IJRES)*, 6(2), 495-503. <https://doi.org/10.46328/ijres.v6i2.1110>

Ally, M. (2009). Mobile learning: Transforming the delivery of education and training. *Athabasca University Press*.

Gikandi, J. W., Morrow, D., & Davis, N. E. (2011). Online formative assessment in higher education: A review of the literature. *Computers & Education*, 57(4), 2333-2351. <https://doi.org/10.1016/j.compedu.2011.06.004>

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Schmid, R. F., Narayanan, N. H., & Eguchi, A. (2019). The impact of the quality of electronic resources on student performance: A comparative study between two university courses. *Education and Information Technologies*, 24(5), 2779-2800. <https://doi.org/10.1007/s10639-019-10076-2>

Somyürek, S. (2017). The challenges of technology integration into mathematics education from the perspective of computer teachers: A literature review. *EURASIA Journal of Mathematics, Science and Technology Education*, 13(2), 371-391. <https://doi.org/10.12973/eurasia.2017.00647a>

(20) Capsule Curriculum Vitae (One-page brief C.V. only)

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Academic History

Master of Engineering, Western Institute of Technology, 2019

Master of Science in Computer Science (CAR), Iloilo Science and Technology University, 2013

Tertiary AMA University, QC Campus, 1998, Bachelor of Science in Computer Engineering 1998

Professional Experience:

- Assistant Professor IV
Capiz State University, Dayao Satellite College
2023 to date
- Department Chairperson
BSCS Department, Capiz State University, Dayao Satellite College
2000-2002, 2004-2017, 2023-2024 (1st Semester only)
- Research Coordinator
BSCS Department (current)
- Extension Coordinator
BSCS Department, Capiz State University, Dayao Satellite College
2005 - 2022